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S.I.: DATA MINING AND DECISION ANALYTICS

When to declare the third innings of a test cricket match?

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Abstract
When to declare the third innings in a test cricket match is a crucial decision directly impacting the outcome of the match. The captain of the side batting in the third innings takes into account factors like lead runs, batting strength of the opposition, favorability of the pitch, approximate number of overs left in the game, etc. to make the decision. The objective of this study is to develop a decision support system for the captain using machine learning algorithms to predict the outcome of a test match at different stages of the match. This will aid the captain to decide when to declare. Several new crucial factors are identified that affect the match outcome. Declaration decisions of past test matches are analyzed using probability functions of win, loss, and draw derived using these models. Previous researches have used only simple regression based techniques to predict the match outcome with low accuracy. Data of 354 test matches from 2008 to 2017 has been used to train and test the algorithms. Support vector machine is found to be the most accurate with an accuracy of 88.8%.

Keywords Test cricket · Neural network · Support vector machines · Extreme gradient boosting

1 Introduction

Test cricket is the longest and one of the oldest forms of cricket. It is played between two teams each consisting of 11 players. In the course of a test match, maximum of four innings are played—at most two by each team—with a maximum time interval of five days. One team bats and the other bowls during each innings. Coin toss decides which team begins the batting/bowling. There are various ways in which a team's innings can end. The important ones are as follows:

- The team is all out while batting, i.e., the team has lost 10 out of 11 batsmen/wickets.
- The captain closes/declares the batting innings whenever he/she feels that the team has enough runs to win or draw the match.

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– The team batting in the fourth innings scores the required runs to win the match.
– The prescribed time (five days) expires and the match ends in a draw.

It is not necessary that the team that bats in the first innings bats again in the third. This is possible if the opponent could not overcome or trails by more than 200 runs compared to the first team's runs scored in the first innings. Under such circumstances, the first team can enforce the trailing team to have their second innings of batting next. This phenomenon is known as *follow-on*. If the trailing team could not overcome the runs of the first team after two innings of batting, the match terminates in only three innings without requiring the first team to bat again and is declared the winner. Otherwise, the first team bats again in the fourth innings. The match may terminate in fewer than four innings as well if the prescribed time of five days is over. In that case, the match will end in a draw.

In yet another situation, the team batting in the third innings may think they have sufficient runs to overwhelm the opponent team, terminates its batting before losing 10 wickets (known as *declaration*), and allows the opposition to commence their next innings batting. The decision of the captain to declare the third innings may drastically affect the course of the match. An early decision can lead to a defeat and a late declaration might draw the match. The probability of win keeps increasing and loss is decreasing with every run added to the lead till a certain point after which the probability of a draw dominates. The point of declaration depends on the nature of the captain. A risk-loving/aggressive captain would offer higher weightage to win than draw/loss and might declare the third innings early. On the other hand, a risk-averse/conservative captain would try to minimize the chances of loss.

The analysis presented in this paper is considered from the perspective of the team batting in the third innings. We focus on batting in the third innings as it is a strategic part of the match, unlike the first two innings where each team attempts to score maximum runs and the final innings where a team tries to either win or draw the match. To decide if it is suitable to declare the third innings at a particular point requires a captain to consider various factors like lead runs, previous batting performance of the opponent, their bowling strength, nature of the entire game played till now, weather conditions, pitch condition, overs remaining, etc. In this study, several factors that affect match outcome (described in Sect. 3.2) have been considered to develop machine learning models based on historical data and predict the match outcome due to third innings declaration.

The paper is organized as follows. The next section summarizes the literature on similar studies for the test cricket match. Section 3 describes the data, attributes used, basic statistical analysis, and preprocessing steps. Machine learning based models and computational results using them are presented in Sect. 4. Declaration decisions of past matches are analyzed in Sect. 5. Finally, the paper ends with conclusions and directions for future research in Sect. 6.

2 Related work

Few researches have been carried out in the past on the prediction of match outcomes in test cricket. Brooks et al. (2002) developed an ordered response model for prediction of test cricket outcome for the period 1994 to 1999 and concluded that the outcome can be explained by the team's batting and bowling strengths. The independent variables considered were categorized as batting measures (runs per wicket and runs per over) and bowling measures (runs per wicket and runs per over of the opposition). The accuracy of the model was 71%, but poor accuracy for draw prediction (57%). The authors also performed a 'style' analysis of the teams which showed all the teams can be categorized under five unique styles.

Multinomial logistic regression based model has been used by Scarf and Shi (2005) to predict match outcome and probabilities of win, draw and loss at potential declaration points of third innings. The idea is further broadened to compute probabilities of match outcome based on target aimed for and run-rate. This work has been extended by Scarf and Akhtar (2011) to model match outcomes at the end of first, second and third innings. They analyze how declaration decisions vary from innings to innings and found overs remaining and lead runs to be important factors. The authors unfold the work further and predict the match outcome session by session in Akhtar and Scarf (2012). They categorize the factors affecting outcome as—"pre-match effects" (which includes the strength of teams, ground effect, home field advantage, outcome of the toss) and "in-play effects" (lead runs, overs-remaining, run-rate, and wicket remaining). Pre-match effects are found to be good predictors of match outcome early on, however, the in-play effects dominate later. Scarf et al. (2011) find the run rate to be adopted in third innings and target to be set for the opponent to maximize the probability of a win. They consider run rate as a substitute for batting strategy and study the influence of run rate on the probability of outcomes. Also, the authors model the runs scored in the third innings as a negative binomial distribution.

Another use of the multinomial logistic regression model can be seen in Allsopp and Clarke (2004) to quantify the relative batting and bowling strengths of a team playing one-day cricket or test cricket. The authors also explore how other features—home advantage, winning the toss, lead in first innings—affect the outcome of test matches. A team's first innings batting and bowling strength, first innings lead, batting order, and home advantage are found to be strong predictors of a win. Least square regression has been used by Allsopp et al. (2002) to predict first innings score and outcome of a test cricket match. First innings lead and duration (in minutes), and home advantage are found to be significant factors. According to the authors, the home team should establish a lead in excess of 93 runs on an average to improve chances of winning, whereas it is 115 runs for the away team. Moreover, there is a higher chance of draw if an innings extends beyond 277 overs. Perera et al. (2014) provide third innings declaration guidelines considering two new factors—desire to win versus draw and scoring characteristics of the match.

Some researchers observed batting and bowling averages are not sufficient to evaluate a player's performance. Akhtar et al. (2015) propose a new player rating system for test cricket. They compute match outcome probabilities session by session and utilize them to measure batting, bowling and fielding contributions of each player. This assessment can be useful to determine the 'best' player in a match, a series or a calendar year. A study to select players considering not only the player's efficiency, but also its consistency, and fit with the team strategy using data envelopment analysis method can be found in Adhikari et al. (2018). Stevenson and Brewer (2017) present a Bayesian approach to model the batting ability of batsmen in test cricket.

Besides statistical techniques, machine learning algorithms also have been applied to cricket. Neural networks based model has been developed by Iyer and Sharda (2009) to select players for international tournaments like the World Cup. The authors predict the future performance of the cricketers based on their past. Cricketers are categorized as performer, moderate, and failure. Their analysis was based on data from 1985 till 2007. The reader is referred to Schumaker et al. (2010); Haghighat et al. (2013) for a review on data mining techniques for building decision support systems to predict results of a game, performance assessments of different players, developing game strategy by observing patterns, team selection, etc. These surveys are not limited to cricket.

The majority of the papers mentioned above predict the outcome of a test cricket match using multinomial logistic regression which is linear in nature and using too obvious factors.

Moreover, there is limited literature on finding the best declaration point in the third innings. The contributions of our paper are multifold. (1) We have addressed the problem of finding the match outcome using advanced machine learning algorithms such that non-linear relationships between factors can also be captured, leading to improved accuracy. (2) Several new factors that are intrinsic to match status have been identified which explain the variance in match outcome much better. The importance of various factors in predicting the outcomes of a match and the effect of their variation on probabilities of the outcomes have been investigated. (3) Additionally, the declaration decision of past test matches has been analyzed with respect to the probabilities of win, loss and draw generated from our models.

3 Solution methodology

The section describes the steps undertaken as a precursor to building the prediction models. It consists of describing the data, identification of crucial attributes for outcome prediction, understanding these attributes using basic statistical techniques, and preparing the data for training the models. Each of these steps is described in detail in the following subsections.

3.1 Data description

Data on 403 international test matches from Jan 02, 2008 to Jul 06, 2017 was obtained from Cricsheet website (2019). This website contains the ball-by-ball score of each international test match. Matches in which "follow-on" was forced (39 matches) were ignored from our analysis as the team enforcing follow on never lost such matches. Ten more matches were removed from consideration as they were abandoned due to weather conditions without a single ball being bowled. We do not remove any other match affected due to poor weather unlike other researches (Akhtar and Scarf 2012) in the past. Although this may adversely affect the accuracy of our models due to the unpredictable nature of such matches, we cannot remove them due to their practical relevance.

Out of the remaining 354 matches, with respect to the team batting in the third innings, 146 matches were won, 130 were lost, and 78 were a draw. The third inning was declared in 170 matches. Year wise distribution of the matches is given in Table 1. Performance of the 10 test playing countries while batting first is shown in Table 2. Note that this is not the same as the total wins, losses and draws by the team.

Winning the toss is generally considered to be crucial for the match to go in favor of the team winning it. Distribution of the match outcome with the toss decision is presented in Table 3. Most of the teams chose to bat first on winning the toss. Of course, this decision also depends on several other factors, such as pitch conditions, etc. The number of matches won by batting first (42.6%) is slightly higher than those bowling first (38.2%). However, these wins cannot be attributed to only winning the toss. A study by De Silva and Swartz (1997) has found wining the coin toss does not provide any competitive advantage.

For training and testing the machine learning algorithms, the dataset is split into 70:30 ratio respectively, such that the training set consisted of 245 matches and the test set of 109 matches. Since our models have to predict the outcome of future matches, they are trained on matches from Jan 02, 2008 till Aug 15, 2014 to predict the outcome of matches from Sep 14, 2014 till Jul 06, 2017.

Table 1 Year wise distribution of matches

Year	Win	Loss	Draw	Total
2008	12	18	8	38
2009	14	8	11	33
2010	11	15	8	34
2011	12	13	12	37
2012	11	17	9	37
2013	13	17	10	40
2014	19	12	7	38
2015	21	11	6	38
2016	24	13	4	41
2017	9	6	3	18
Total	146	130	78	354

Table 2 Team wise performance while batting first

Team	Number of matches			Probability of		
	Won	Lost	Draw	Win	Loss	Draw
Australia	34	18	13	0.52	0.28	0.20
Bangladesh	5	13	3	0.24	0.62	0.14
England	21	16	13	0.42	0.32	0.26
India	20	12	9	0.49	0.29	0.22
New Zealand	14	17	8	0.36	0.44	0.21
Pakistan	15	11	5	0.48	0.35	0.16
South Africa	11	8	12	0.35	0.26	0.39
Sri Lanka	18	14	11	0.42	0.33	0.26
West Indies	5	15	4	0.21	0.63	0.17
Zimbabwe	3	6	0	0.33	0.67	0.00

Table 3 Distribution of match outcome with toss decision

Toss decision	Win	Loss	Draw	Total
Bat	104	86	54	244
Bowl	42	44	24	110

3.2 Feature extraction

Raw data consists of attributes/variables such as innings wise score (runs, wickets and overs), ground, and date of the match. The data extracted from the website was not too meaningful for predicting the outcome of the match. Some additional factors on which the declaration decision or match outcome may depend are proposed as follows. A brief explanation of them is also provided.

1. Team batting in first innings (T1)—one of the 10 international teams playing test cricket.
2. Team bowling in first innings (T2)—one of the 10 international teams playing test cricket.

3. Team winning the toss (TW)—one of the 10 international teams playing test cricket.
4. Toss decision (TD)—the decision taken to bat or bowl by the toss winning team.
5. Ground Country (GC)—country of the ground where the match is played. This variable is considered instead of the usual binary variable "Home Ground" to take care of neutral venues.
6. Month (MN)—the month of the year.
7. Follow On (FO) = 1 if follow on was imposed by the team batting in the first innings to the opponent, = 0 otherwise.
8. Sudden fall of wickets in the first innings (SW1) = 1 if three or more wickets have fallen within 30 runs or in 20 overs in the first innings, = 0 otherwise.
9. Sudden fall of wickets in the second innings (SW2)—similar to SW1 for second innings.
10. Sudden fall of wickets in the third innings (SW3)—similar to SW1 for third innings.
11. Match result (MR)—class label/ target attribute denoting the outcome of the match—win, loss, draw.
12. Strength of team T1 (TS1)—measured by the fraction of matches won by team T1 in past two years.
13. Strength of team T2 (TS2)—measured by the fraction of matches won by team T2 in past two years.
14. Batting strength of team T1 (T1BT)—it is the average batting rating of the team using the batting rating of individual players for the last six months provided by ICC.
15. Batting strength of team T2 (T2BT)—similar to T1BT for team T2.
16. Bowling strength of team T1 (T1BW)—it is the average bowling rating of the team using the bowling rating of individual players for the last six months provided by ICC.
17. Bowling strength of team T2 (T2BW)—similar to T1BW for team T2.
18. Runs scored in first innings (R1).
19. Runs scored in second innings (R2).
20. Runs scored in third innings (R3).
21. Over bowled in first innings (O1).
22. Over bowled in second innings (O2).
23. Over bowled in third innings (O3).
24. Wickets fallen in first innings (W1).
25. Wickets fallen in second innings (W2).
26. Wickets fallen in third innings (W3).
27. Batting strength in first innings (BT1) = 1 if any batsman scores 70 runs or more, or any partnership greater than 100 runs, = 0 otherwise.
28. Batting strength in second innings (BT2)—similar to BT1 for second innings.
29. Batting strength in third innings (BT3)—similar to BT1 for third innings.
30. Bowling strength in first innings (BW1) = 1 for any five-wicket haul by any bowler in first innings, = 0 otherwise.
31. Bowling strength in second innings (BW2)—similar to BW1 for second innings.
32. Bowling strength in third innings (BW3)—similar to BW1 for third innings.
33. Team 1 dominance (T1D)—if team T1 has high lead runs at the end of the second innings and less number of overs being bowled in the first two innings, team T1 has high probability of winning the match. We consider lead runs of more than 150 at the end of the second innings and less than 250 overs bowled in the first two innings as an indicator of team T1's dominance.

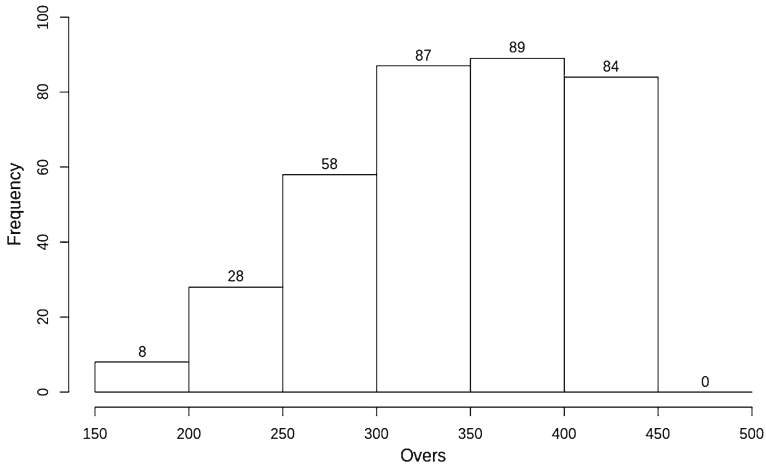


Fig. 1 Distribution of total overs

34. Team 2 dominance (T2D)—if team T1 has a high run deficit at the end of second innings, team T2 has higher chances of winning the match. We consider team T2 to be dominating if team T1 trails by more than 200 runs at the end of the second innings.
35. No dominance (ND)—if high overs have been bowled in the first two innings and lead runs for team T1 at the end of the second innings is low, none of the teams dominate, and most likely the match will end in a draw. This is because team T1 will consume a lot of overs in the third innings to achieve a reasonable lead, thereby leaving very few overs for the fourth innings. We consider less than 150 runs as the lead for team T1 at the end of the second innings and more than 300 overs bowled in the first two innings as indicators of no dominance.

Note that variables like overs remaining and lead runs are not used here unlike most of the work in the literature. Instead, variables like O1, O2, O3 and R1, R2, R3 have been used. This is because the concept of overs remaining considers the match will end at 450 overs. The model will assume the number of overs in the fourth innings to be intact even if time is lost due to bad weather. Rarely 450 overs have been bowled in test matches. Moreover, overs remaining and lead runs are derived variables with respect to O1, O2, O3, R1, R2 and R3, and therefore may result in loss of important information about the status of the match which is beyond our comprehension. Using actual values of O1, O2 and O3 instead of overs remaining do not require information of overs lost due to bad weather, unlike previous studies. A histogram of total overs bowled to substantiate the above argument is shown in Fig. 1. The highest number of overs bowled in a match in the dataset is 448 (in only two matches) and above 430 overs in only 23 (Fig. 2) out of 354 matches.

To measure the strength of a team, we have considered two types of variables. Variables TS1 and TS2 measure the strength of a team as a single unit. On the other hand, variables T1BT, T2BT, T1BW, T2BW account for the team strength considering the ICC rating of each player composing the team. Defining the variables in this new way is a crucial contribution to our paper.

Three categories of models—M0, M2 and M3 are built using these variables which predict the outcome of the match at different stages. Model M0 predicts the match outcome before the match starts using "pre-match" attributes. Models M2 and M3 are used to predict the match

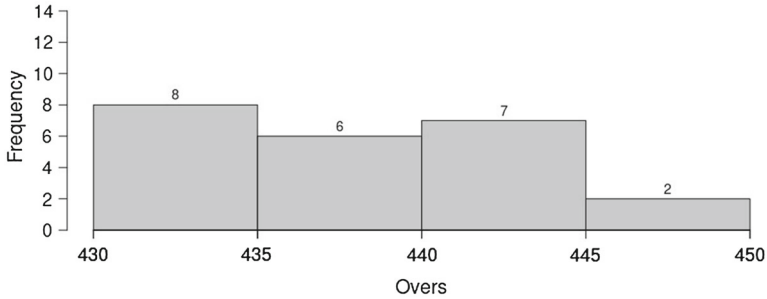


Fig. 2 Distribution of total overs between 430 and 450

Table 4 Categorical variables considered for different models

Attribute	Levels	M0	M2	M3
T1	10	✓	✓	✓
T2	10	✓	✓	✓
TW	10	✓	✓	✓
TD	2	✓	✓	✓
GC	13	✓	✓	✓
MN	12	✓	✓	✓
FO	2		✓	
SW1	2		✓	✓
SW2	2		✓	✓
SW3	2			✓
BT1	2		✓	✓
BT2	2		✓	✓
BT3	2			✓
BW1	2		✓	✓
BW2	2		✓	✓
BW3	2			✓
T1D	2		✓	✓
T2D	2		✓	✓
ND	2		✓	✓
MR	3	✓	✓	✓

outcome at the end of second and third innings respectively. A summary of the categorical variables is presented in Table 4 and that of numeric variables in Table 5. Basic statistical quantities like mean, standard deviation, quartiles, minimum and maximum computed for numeric variables such as O1, O2, O3, R1, R2 and R3 are also reported. The checkmarks in these tables indicate the attributes used in models M0, M2 and M3.

3.3 Exploratory data analysis

In this section, we describe the effect of different variables on match outcome using their conditional density (CD) plots. These plots are powerful in examining how the conditional

Table 5 Numeric variables considered for different models

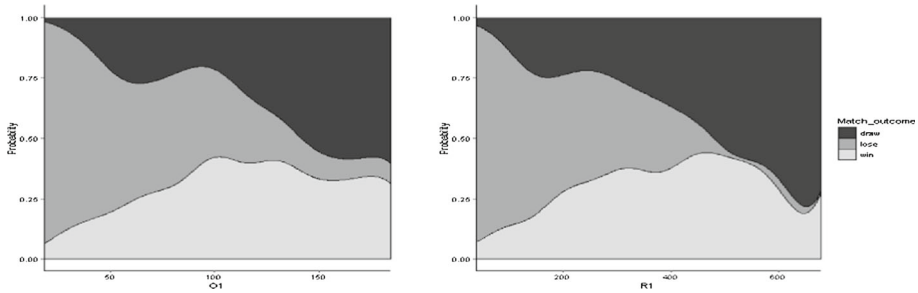
Attribute	μ	σ	Min	Q1	Q2	Q3	Max	M0	M2	M3
TS1	0.6	0.2	0	0.5	0.6	0.7	1	✓	✓	✓
TS2	0.5	0.2	0	0.5	0.5	0.6	1	✓	✓	✓
T1BT	0.6	0.1	0	0.5	0.6	0.7	1	✓	✓	✓
T2BT	0.4	0.2	0	0.3	0.4	0.5	1	✓	✓	✓
T1BW	0.6	0.2	0	0.5	0.6	0.7	1	✓	✓	✓
T2BW	0.5	0.2	0	0.3	0.5	0.6	1	✓	✓	✓
R1	338.8	135.5	7	236	325	444	679		✓	✓
O1	106.1	36.4	1.4	77.05	107.2	134.6	184.1		✓	✓
W1	9.3	1.5	2	10	10	10	10		✓	✓
R2	318.6	143.9	15	211.5	299	412	752		✓	✓
O2	104.0	41.9	11.6	70.0	100.2	134.3	248.5		✓	✓
W2	9.6	1.4	0.0	10.0	10.0	10.0	10.0		✓	✓
R3	246.4	98.5	23.0	172.0	238.0	302.0	650.0			✓
O3	80.5	33.2	14.6	56.6	77.6	99.6	209.6			✓
W3	8.2	2.6	0.0	6.0	10.0	10.0	10.0			✓

distribution of a categorical variable (match outcome here) changes with some continuous/numeric variable (Hogg et al. 2005). Suppose random variables X and Y have a jointly continuous distribution with joint density $f(x, y)$. The conditional probability density function of Y given the occurrence of the value x of X is written as

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{\text{joint } (X, Y) \text{ density at } (x, y)}{\text{marginal } X \text{ density at } x}$$

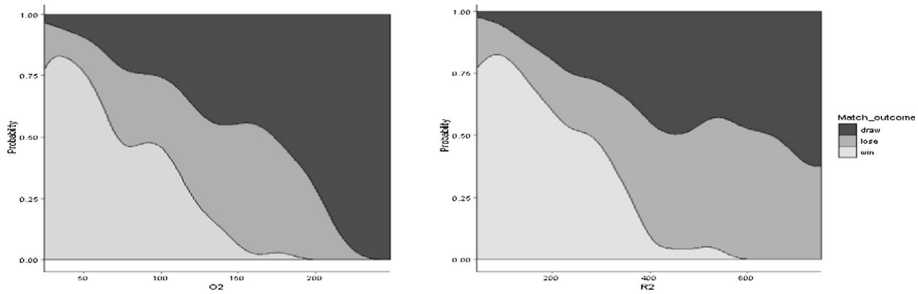
Four sets of CD plots are analyzed as follows with respect to team T1. These plots have been crucial for deciding the threshold for the variables T1D, T2D and ND.

1. *CD plot of O1 and R1* (Fig. 3): As the overs and runs in the first innings increase, probability of loss decreases, probability of win increases till around 430 runs or 120 overs then decreases and finally the probability of draw dominates.
2. *CD plot of O2 and R2* (Fig. 4): Chances of win constantly decrease if the opposition bats too long and scores high runs in the second innings. However, the match tends to draw if the second innings continues beyond around 130 overs or 430 runs.
3. *CD plot of O3 and R3* (Fig. 5): If team T1 bats for less than 90 overs or 300 runs in the third inning, a loss is highly likely. However, beyond 130 overs or 380 runs it is highly probable that the match ends in a draw. This clearly indicates the need to identify the optimal value of R3 for the declaration of the third innings.
4. *CD plot of LR3* (Fig. 6): As the lead runs in the third inning (LR3) increases beyond 250, the probability of losing the match decreases and that of win increases, which is expected. However, too high LR3 may result in the match to end in a draw. The observation of the probability of draw from the figure is contradictory, i.e., the probability of the match being drawn first increases then decreases. This is because in the matches in which LR3 was too high the opposition was too weak which resulted in a win for the team.



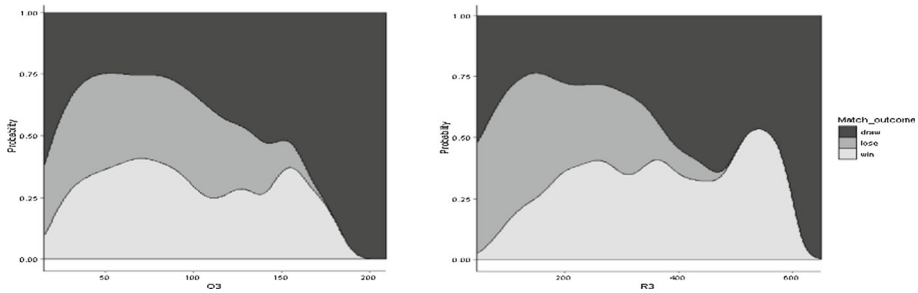
(a) O1 CD plot **(b)** R1 CD plot

Fig. 3 Conditional density plot of O1 and R1



(a) O2 CD plot **(b)** R2 CD plot

Fig. 4 Conditional density plot of O2 and R2



(a) O3 CD plot **(b)** R3 CD plot

Fig. 5 Conditional density plot of O3 and R3

3.4 Data preprocessing

Rigorous data preprocessing was not required as there was neither an outlier nor missing values in the data downloaded from the Cricsheet website (2019). However, it was necessary to identify important factors for prediction as many potential factors have been proposed in Sect. 3.2. The importance of the factors is accessed using random forest importance of variables. The importance of the variables is presented in Fig. 7. For predicting draw and

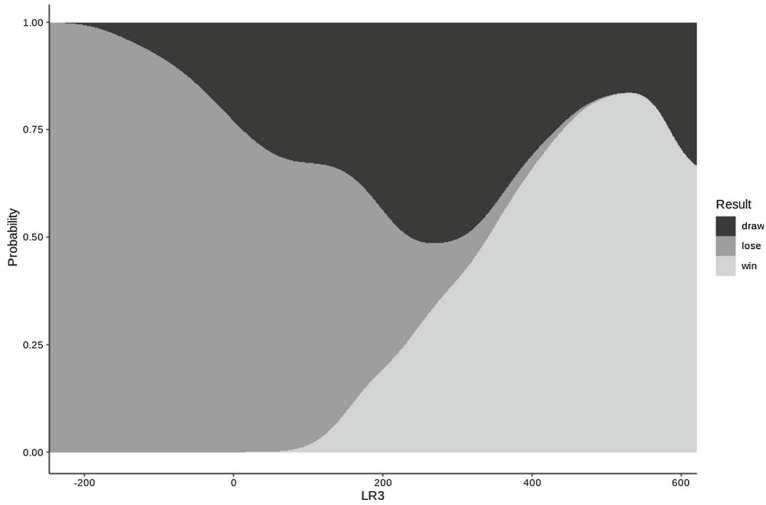


Fig. 6 Conditional density plot of LR3

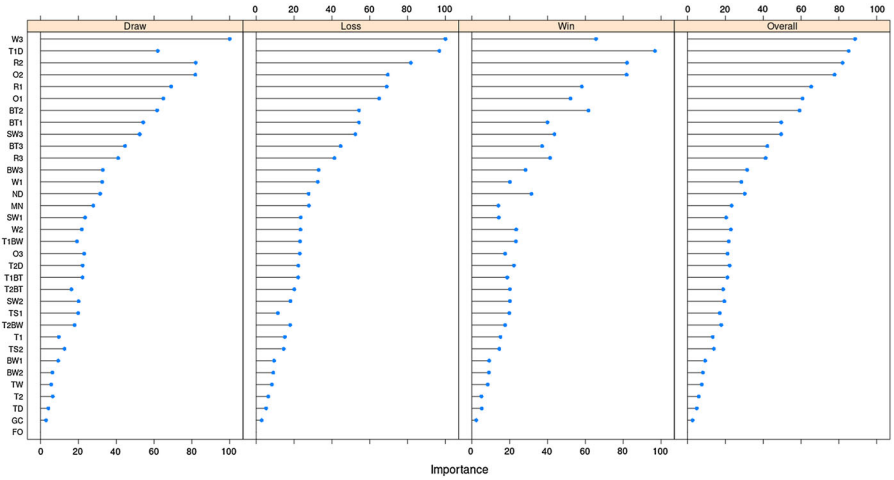


Fig. 7 Importance of variables using random forest

loss, variable W3 has the highest importance. Variable T1D has the highest significance for predicting a win. This is as expected, and thus validates our models.

4 Model building

This section discusses in detail the building of models M0, M2 and M3 using machine learning algorithms. Variables identified as important for prediction during pre-processing are fed to the algorithms as input. k-fold cross-validation has been used to estimate the accuracy of the algorithms and tune their parameters. Libraries available in R programming language has been used to implement the models.

Table 6 Confusion matrix using ANN for M3

	Actual		
	Win	Loss	Draw
Predicted			
Win	51	1	2
Loss	2	30	2
Draw	7	3	9

Table 7 Confusion matrix using SVM for M3

	Actual		
	Win	Loss	Draw
Predicted			
Win	54	1	2
Loss	5	32	2
Draw	1	1	9

4.1 Cross-validation

We have applied k-fold cross-validation (Kohavi 1995) to assess the goodness of fit of all the machine learning algorithms while varying their tuning parameters. The advantage of using this technique is that it gives an estimate of the accuracy of a model even before testing it on test data. We can iterate it over various tuning parameters to choose the values that optimize the model.

4.2 Artificial neural networks (ANN)

An artificial neural network (Hagan et al. 1996) has been trained using "keras" and "tensorflow" packages. The network consists of 16 nodes in the input layer, 8 nodes in each of the two hidden layers and three nodes in the output layer. A dropout layer is used to avoid overfitting with a dropout rate = 0.2. Softmax activation function is used for the nodes at the output layer whereas rectified linear activation function for nodes in the rest of the layers. The accuracy achieved is 45.5% for M0, 74.6% for M2 and 84.1% for M3 on the test data. The confusion matrix obtained using ANN for model M3 is shown in Table 6.

4.3 Support vector machines (SVM)

A support vector machine (Cortes and Vapnik 1995; Fung and Mangasarian 2005) was trained using "e1071" package having method="C-classification", kernal="sigmoid", and tuning parameters γ = 0.004 and misclassification cost C = 1.0. The algorithm resulted in accuracy of 44.8% for M0, 76.3% for M2 and 88.8% for M3. Confusion matrix obtained using SVM for model M3 is presented in Table 7.

Table 8 Confusion matrix using XGboost for M3

	Actual		
	Win	Loss	Draw
Predicted			
Win	39	2	2
Loss	8	32	3
Draw	13	0	8

Table 9 Confusion matrix using MLR for M3

	Actual		
	Win	Loss	Draw
Predicted			
Win	36	3	2
Loss	9	30	2
Draw	15	1	9

4.4 Extreme gradient boosting (XGboost)

This is a relatively new method where machine learning techniques are implemented under gradient boosting framework (Chen and Guestrin 2016). New models are added subsequently to correct the error caused by the previous model. This process is continued till no further improvement is possible. To minimize the loss caused by merging two models it deploys the gradient descent technique.

Package "e1071" was used to train the XGboost algorithm with tuning parameters nrounds = 1000, max_depth = 3, $\eta = 0.001$, $\gamma = 0.1$, colsample_bytree = 1, and subsample = 1. Accuracy obtained is 44.6% for M0, 66.1% for M2 and 73.8% for M3. Table 8 shows the confusion matrix obtained using XGboost for model M3.

4.5 Multinomial logistic regression (MLR)

This is a generalization of the logistic regression to multiclass problems (Böhning 1992). The algorithm is trained using "nnet" package available in R with default parameters. The accuracy achieved is 43.8% for M0, 59.3% for M2 and 70.1% for M3. Table 9 is the confusion matrix obtained using this technique for model M3.

4.6 Random forest (RF)

It is an ensemble algorithm that creates a set of decision trees from randomly selected subsets of training set (Breiman 2001). By using the "randomForest" function available in R, and tuning parameter mtry = 15, ntree = 25, the best accuracy was achieved as 54.2% for model M0, 70.3% for model M2 and 80.4% for model M3. Table 10 is the confusion matrix for model M3.

The accuracy of all the algorithms for the different models is summarized in Table 11. The accuracy for class wise prediction, essentially *sensitivity*, of the algorithms for each model is also presented in this table. From the pictorial view of the overall accuracy of all the models

Table 10 Confusion matrix using Random Forest for M3

	Actual		
	Win	Loss	Draw
Predicted			
Win	44	1	1
Loss	4	31	1
Draw	12	2	11

Table 11 Accuracy of the models

Algorithm	Overall accuracy			Win accuracy			Loss accuracy			Draw accuracy		
	M0	M2	M3	M0	M2	M3	M0	M2	M3	M0	M2	M3
ANN	45.5	74.6	84.1	52.9	81.4	85.0	44.1	82.4	88.2	17.7	21.4	69.2
RF	54.2	70.3	80.4	54.3	71.4	73.3	58.8	76.5	91.2	43.8	50.0	84.6
SVM	44.8	76.3	88.8	36.7	87.1	90.0	65.5	64.7	94.1	37.5	50.0	69.2
XGB	44.6	66.1	73.8	47.1	70.0	73.8	55.9	64.7	94.1	11.8	50.0	61.5
MLR	43.8	59.3	70.1	41.4	64.3	70.1	58.8	52.9	88.2	23.5	50.0	69.2

in Fig. 8, it is easy to observe that accuracy of the algorithms increases significantly due to the addition of more relevant variables. Additionally, the prediction of the outcome is much better as the match progresses. Prediction using SVM dominates all the other algorithms for models M2 and M3. SVM is found to be the most accurate with the overall accuracy of 88.8% under model M3. None of the researches in the literature reported so high accuracy to the best of our knowledge. Since most of the models proposed in the literature are based on linear regression, the authors have reported R^2 value. However, the percentage of correctly predicted matches has been reported in Akhtar and Scarf (2012) and Brooks et al. (2002), which is the same as accuracy. The analysis carried out by Akhtar and Scarf (2012) is from the perspective of predicting match outcome, not from a declaration point of view, at different stages of the match. Generally, the third innings declaration happens around lunch of Day 4. Although we cannot compare our models exactly with the ones in Akhtar and Scarf (2012), accuracy reported by Akhtar and Scarf (2012) is around 78–80% during start—lunch of Day 4. Authors in Brooks et al. (2002) have reported the following accuracies - 74.8% for win, 57.1% for draw, 80.9% for loss and 71.1% overall.

From the sensitivity measures (provided in Table 11 and the confusion matrices) it is clear that SVM is best for win prediction, SVM and XGboost are equally good for loss prediction, and RF for draw prediction. Loss prediction has the highest accuracy (94.1%) followed by win prediction (90.0%) and then draw prediction (84.6%). Our models have been tuned to predict loss more accurately than a win or draw as the misclassification cost is very high for this case. Although it is inherently difficult to predict a draw, pointed in Brooks et al. (2002) as well, the models have reasonable accuracy for predicting a draw.

5 Analysis of past declaration decisions

Analyzing the declaration decision taken in past test matches using our proposed models is an interesting contribution of this paper. As pointed out earlier from the CD plot of third innings lead runs, LR3 (Fig. 6), with the delay of declaration, the probability of win decreases

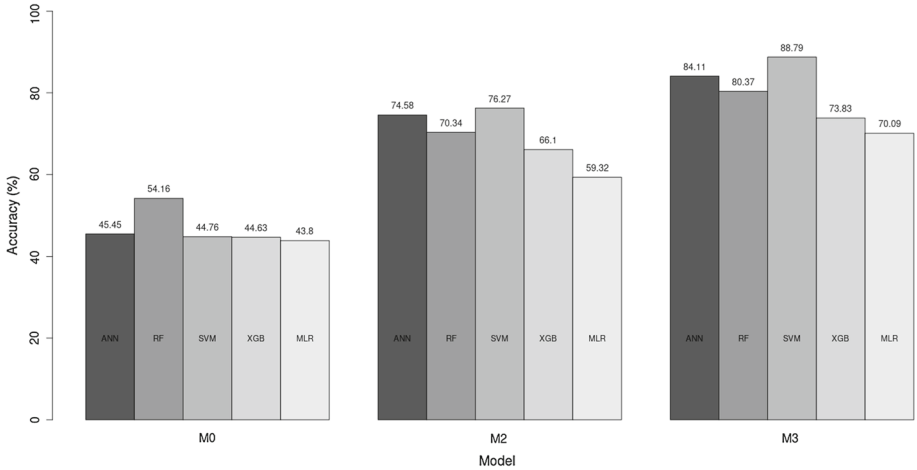


Fig. 8 Overall accuracy of the models at different stages of the match

and that of draw increases. It is ideal to declare the innings at the peak of win probability. However, a captain with conservative nature may continue the third innings till the 10 wickets are lost.

The following four matches are analyzed to verify whether they were declared at the optimal point where win probability was maximum. To do so the third innings was extrapolated assuming the same run rate for 50–100 overs further till the expected fall of 10 wickets in the third innings. The actual declaration point is compared with the ideal one proposed by the models. The probability functions of win, loss and draw with respect to LR3 are shown for each match. The rightmost vertical (dashed) line in each plot depicts the expected end of the third innings when all the wickets fall. This serves as a proxy to know the declaration point if the captain is conservative.

5.1 England versus Sri Lanka, June 12–16, 2014

England declared the third innings at the score of 267/8, setting a target of 390 runs for Sri Lanka to win the match (Table 12). During the fourth innings, Sri Lanka lost 9 wickets at the score of 201. The match ended in a draw as the time was over. From the probability graphs (Fig. 9), we can infer that England made the declaration slightly late. The win probability was maximum when lead runs were around 330 runs, which could have been achieved at around 54th over of the third innings considering the same run rate of third innings. This would have added 15 overs to the fourth innings. From the second innings it can be found that Sri Lanka consumed about 14 overs per wicket on an average. Thus, England could have bowled the last wicket of Sri Lanka had they declared 15 overs earlier and won the match with high probability.

5.2 South Africa versus Sri Lanka, July 16–20, 2014

South Africa declared the third innings with a score of 206/6, setting a target of 370 runs for Sri Lanka to win the match (Table 13). According to the win probability function (Fig. 10),

Table 12 Innings wise score—Eng versus SL 2014

Innings	Team	Score	Overs
1	England	575/9 d	130.3
2	Sri Lanka	453/10	138.4
3	England	267/8 d	69.0
4	Sri Lanka	201/9	90.0

Table 13 Innings wise score—SA versus SL 2014

Innings	Team	Score	Overs
1	South Africa	455/9 d	166.2
2	Sri Lanka	292/10	104.5
3	South Africa	206/6 d	50.2
4	Sri Lanka	216/10	71.3

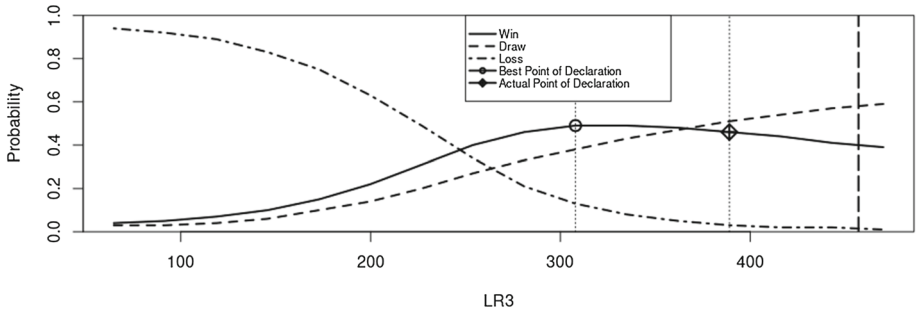


Fig. 9 Win, loss and draw probabilities—Eng versus SL 2014

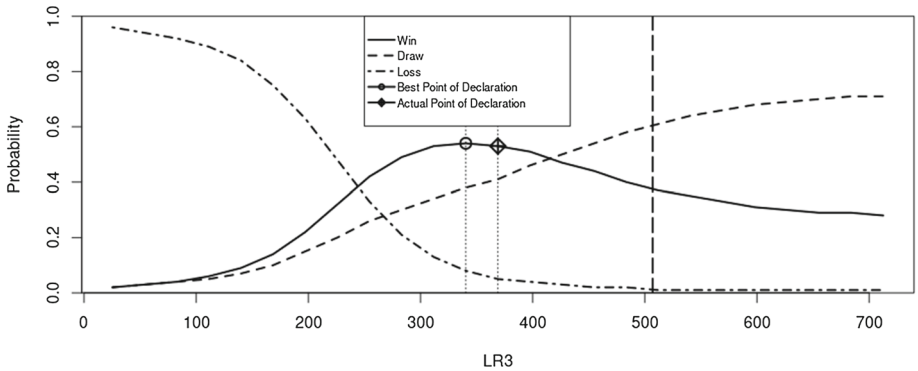


Fig. 10 Win, loss and draw probabilities—SA versus SL 2014

the declaration was not very late compared to the ideal point. Sri Lanka lost all its wicket at a score of 216 runs and South Africa won the match. We can conclude that South Africa made an excellent declaration decision.

Table 14 Innings wise score—SA versus Eng 2012

Innings	Team	Score	Overs
1	South Africa	419/10	139.2
2	England	425/10	126.4
3	South Africa	258/9 d	67.4
4	England	130/4	33.0

Table 15 Innings wise score—Pak versus SL 2012

Innings	Team	Score	Overs
1	Pakistan	551/6 d	147.0
2	Sri Lanka	391/10	124.4
3	Pakistan	100/2 d	18.0
4	Sri Lanka	86/2	22.0

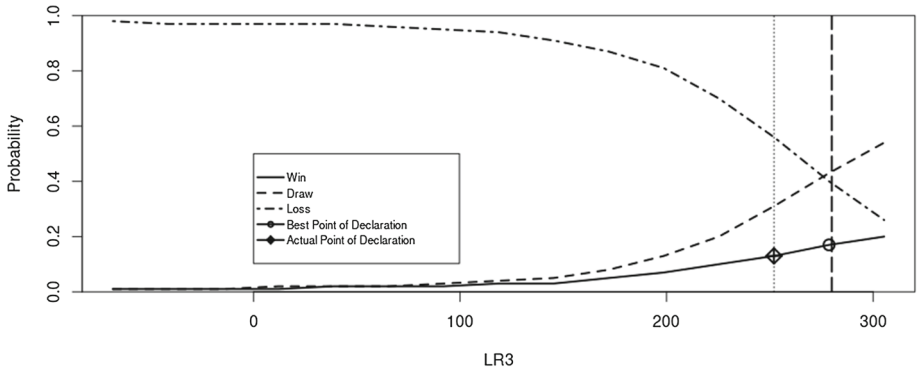


Fig. 11 Win, loss and draw probabilities—SA versus Eng 2012

5.3 South Africa versus England, Aug 02–06, 2012

With a score of 258/9, South Africa declared the third innings of the match, setting a target of 253 runs for England (Table 14). From the probability graphs (Fig. 11), at the point of declaration, the probability to lose the match was higher than that of win and draw. Observing England's performance in the second innings, 253 runs was an achievable target considering about 117 overs remaining in the fourth innings. However, it rained on all five days of the match and caused a huge loss in the playing time. Only 33 overs were bowled in the last innings and 365 overs in total. As a result, the match ended in a draw.

5.4 Pakistan versus Sri Lanka, Jun 30–Jul 04, 2012

Pakistan declared the third innings with a score of 100/2 setting a target of 261 with around 161 overs remaining (Table 15). From the probability graphs (Fig. 12), although the declaration decision was right because the probability to win the match was greater than to draw or loss at this point, the declaration was made way too early. This would have been an achievable target for Sri Lanka given their batting performance in second innings. As per the win probability

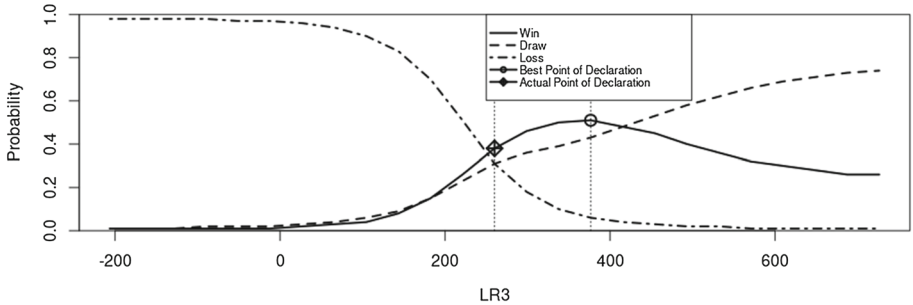


Fig. 12 Win, loss and draw probabilities—Pak versus SL 2012

graph, Pakistan should have set a target of 400 runs. This would have required Pakistan to bat till 43 overs in the third innings assuming the same run rate. Then, Sri Lanka had to score 400 runs in 136 overs (assuming 450 total overs) at the rate of about 3.0 runs per over to win the match. This would have been a competitive target. However, due to rain on two consecutive days of play, only 311 overs were bowled in the match, and the match ended in a draw.

6 Conclusions

In this paper models have been developed using machine learning algorithms to predict the outcome of test matches at different stages of the game. This acts as a decision support system for the captain of the team batting in the third innings to find the best point for declaring the innings. Computational results indicate accuracy as high as 88.8%, significantly higher than those reported in the literature. This is attributed to the identification of new crucial factors such as sudden fall of wickets and better definition of well known factors which sensitively captures the progress of the match. Innings wise batting and bowling strengths have been considered. The limitation of the attribute "overs remaining" has been pointed out. Models have been validated by comparing past declaration decisions.

Condition of pitch changes as the match progresses, but due to practical limitations, this factor has been ignored from model building. However, machine learning models are capable of learning this pattern implicitly from historical matches. Time lost or play affected due to weather conditions or poor light cannot be captured due to the unavailability of data for previous matches. Changes in the team due to on-field injuries are ignored. Although we have contributed significantly towards improving the accuracy of draw prediction, still it remains an important area of research. The current models cannot capture the captain's risk loving nature due to the unavailability of this data. Therefore, the contribution of human behavior cannot be accurately evaluated.

Nowadays, the test matches are getting over in 3-4 days, perhaps it is becoming a faster game. The models we have used cannot accommodate Year as a variable since future values of Year cannot be included in the training data. Therefore, the matches have to be analyzed from a time-series perspective to accommodate this fact.

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